Paper-list quality under a low cap

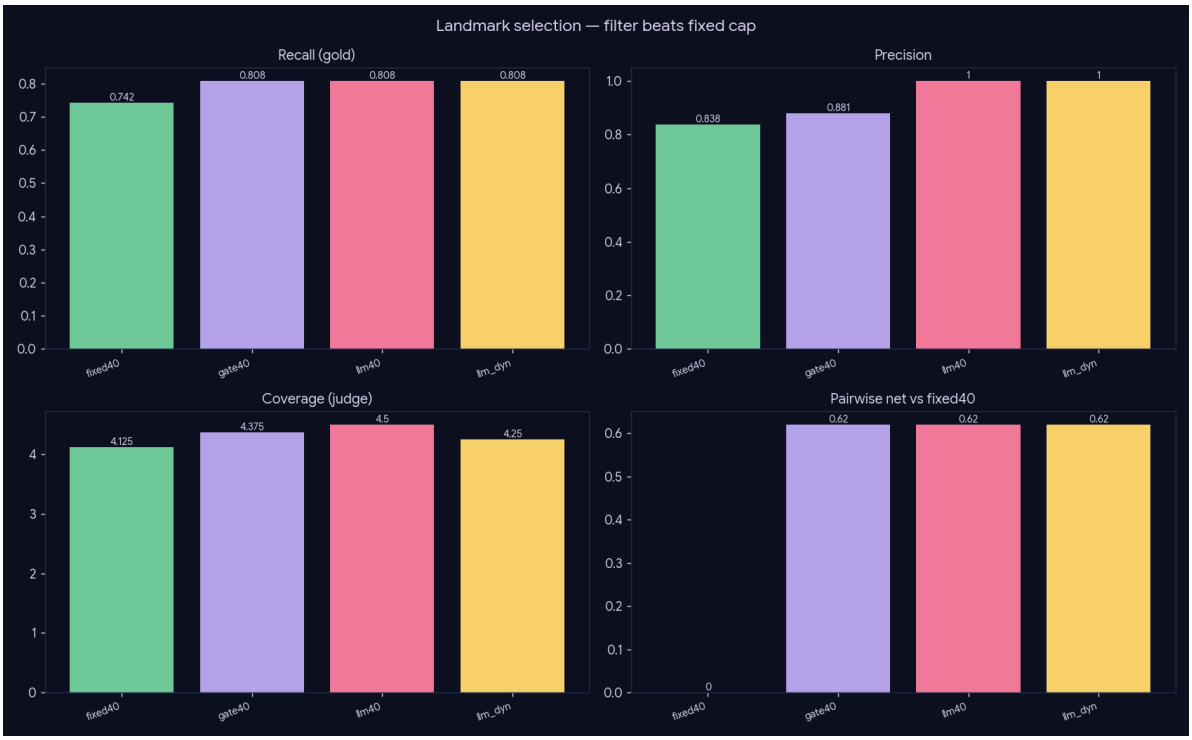
arxivkg · generation · landmark selection

What we did

The committed retrieval takes a hard PPR top-40. Does relevance-filtering that list beat it, and does the cap need to grow? Four arms over one shared pool (same adjacency-list + extracted digest; only the landmark *set* differs), disentangling **filter-type** from **count**: `fixed40` (PPR top-40), `gate40` (0.35 cosine ancestor gate, top-40), `llm40` (drop role=unrelated over PPR-100, capped 40 — same count), `llm_dyn` (same filter, soft cap 60). Scored on the Exp-1-validated metric: direct list metrics (count, recall[gold], precision[role judge], lineage-span) primary + the compose-judge on its trusted axes (grounding, coverage) + pairwise vs `fixed40` (lineage asterisked); judges date-injected.

Results (8 topics, gold-3 for recall)

arm	count	recall	precision	coverage	overall	pair
fixed40	40.0	0.742	0.838	4.125	4.688	0.0
gate40	40.0	0.808	0.881	4.375	4.75	0.62
llm40	40.0	0.808	1.0	4.5	4.75	0.62
llm-dyn	60.0	0.808	1.0	4.25	4.75	0.62



Conclusion

The win is the FILTER, not the count — keep the list at 40 and filter for relevance; do not expand it. All three alternatives beat `fixed40` (recall 0.742→0.808, pairwise +0.62 = 5W/3T/0L). `llm40` **(filter at the same 40) captures the full win** — perfect precision (1.0), full recall (0.808), best coverage (4.5) — at zero size increase. `llm_dyn` **(60) does not beat** `llm40` **(40)** (identical recall/precision/pairwise, slightly lower coverage: cleaner beats bigger). The cheap `gate40` gets ∼80% of the win; the LLM filter adds the precision polish (0.881→1.0). So replace the hard top-40 with a relevance-filtered top-40 — the cheap gate at minimum, the LLM filter for precision-max. Caveat: n=8; the robust signal is the direct metrics, pairwise margins are modest and on validated axes only.